

UNIT-2 AGILE MINI-PROJECT

Student Record and Academic Management Platform

Complete Agile Project Planning & Documentation

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1. Study of Business Problem and Product Vision

The project is a Student Record and Academic Management Platform. The Unit-2 activity translates the product-discovery work into Agile deliverables for managing student records, attendance, academic performance, course registration and reporting.

Business Problem

Educational institutions need a centralized way to maintain student information, course information, attendance, marks, grades, notifications and reports. A unified platform can make these records easier to maintain and reports easier to generate.

Stakeholders

- Students
- Faculty Members
- Academic/Administrative Staff
- Institution Management
- System Administrator

Functional Requirements

- Student registration and profile management
- Student search, update and deletion
- Course creation, updating, assignment and viewing
- Attendance recording, viewing and reporting
- Marks entry, updating, grade calculation and result-sheet generation
- Login, logout, password reset and role-based access control
- Attendance, examination and result notifications
- Student, attendance, performance and course reports

Non-functional Requirements

- Security

- Usability
- Scalability
- Reliability
- Maintainability
- Performance

MVP

The MVP focuses on authentication, student management, course management, attendance, academic performance and essential reporting. Notifications and advanced analytics can be delivered incrementally.

Roadmap

- Release 1: Authentication and project foundation
- Release 2: Student and course management
- Release 3: Attendance and academic performance
- Release 4: Notifications, reporting and improvements

2. Agile Product Vision

Develop a secure and user-friendly Student Record and Academic Management Platform that enables educational institutions to efficiently manage student records, attendance, academic performance, course registration, and reporting through a scalable web-based application.

Objective

Provide a centralized platform for managing academic information and delivering accurate, accessible and timely records.

Primary Users

- Faculty Members
- Students
- Academic/Administrative Staff
- Institution Management
- System Administrator

Business Benefits

- Centralized academic records
- Reduced manual record maintenance
- Faster attendance and performance reporting
- Better access to student information
- Role-based security
- Incremental delivery of useful features

Success Criteria

- Users can register/login and reach the correct dashboard.
- Authorized users can add, search, update and remove student records.
- Faculty can record attendance and generate attendance reports.
- Marks can be entered and grades/results can be generated.
- Authorized users can access only permitted functions.
- Sprint features pass their acceptance criteria.

Constraints

- Limited student-team development time
- Limited development resources
- Need for secure academic data
- Feature prioritization required for MVP

3. Scrum Team Structure

Role	Responsibility	Duties
Product Owner	Product requirements and priorities	Define requirements; prioritize backlog; represent stakeholders; accept completed work
Scrum Master	Agile process facilitation	Facilitate ceremonies; remove impediments; support collaboration; ensure Agile practices
Development Team	Software delivery	Design; code; test; document; demonstrate completed features

4. Project Epics

Epic 1 – Student Management

Purpose: Manage the functions included in this high-level module.

- Student Registration
- Student Profile Management
- Student Search
- Student Update
- Student Deletion

Epic 2 – Course Management

Purpose: Manage the functions included in this high-level module.

- Add Course
- Update Course
- Assign Students
- View Course Details

Epic 3 – Attendance Management

Purpose: Manage the functions included in this high-level module.

- Record Attendance
- View Attendance
- Attendance Reports

Epic 4 – Marks and Academic Performance

Purpose: Manage the functions included in this high-level module.

- Enter Marks
- Update Marks
- Calculate Grades
- Generate Result Sheets

Epic 5 – Authentication and Security

Purpose: Manage the functions included in this high-level module.

- User Login
- Logout
- Password Reset
- Role-Based Access Control

Epic 6 – Notifications

Purpose: Manage the functions included in this high-level module.

- Attendance Alerts
- Examination Notifications
- Result Notifications

Epic 7 – Reporting

Purpose: Manage the functions included in this high-level module.

- Student Reports
- Attendance Reports
- Performance Reports
- Course Reports

5. User Story Repository

ID	Epic	Feature	User Story	Priority	Points
US-01	Authentication	User Login	As a registered user, I want to log in so that I can securely access my dashboard.	High	3
US-02	Authentication	Logout	As a user, I want to log out so that my session is securely ended.	High	1
US-03	Authentication	Password Reset	As a user, I want to reset my password so that I can regain access when I forget it.	Medium	3
US-04	Authentication	Role-Based Access Control	As an administrator, I want role-based access control so that users access only permitted functions.	High	5
US-05	Student Management	Add Student	As an authorized staff member, I want to add a student so that a student record is created.	High	3
US-06	Student Management	Search Student	As an authorized user, I want to search for a student so that I can quickly find a record.	High	2
US-07	Student Management	Update Student	As an authorized user, I want to	High	3

			update student details so that records remain current.		
US-08	Student Management	Delete Student	As an authorized user, I want to delete a student record so that obsolete records can be removed.	Medium	3
US-09	Student Management	View Student Profile	As a user, I want to view a student profile so that relevant academic information is accessible.	High	2
US-10	Course Management	Add Course	As an authorized user, I want to add a course so that it is available for academic management.	Medium	3
US-11	Course Management	Update Course	As an authorized user, I want to update course details so that course information remains accurate.	Medium	2
US-12	Course Management	Assign Students	As an authorized user, I want to assign students to courses so that registrations are maintained.	Medium	5

US-13	Course Management	View Course Details	As a user, I want to view course details so that I can understand course information.	Medium	2
US-14	Attendance	Record Attendance	As a faculty member, I want to record attendance so that attendance records remain accurate and updated.	High	5
US-15	Attendance	View Attendance	As an authorized user, I want to view attendance so that attendance status can be checked.	High	3
US-16	Attendance	Attendance Report	As an authorized user, I want attendance reports so that attendance can be analyzed and shared.	Medium	5
US-17	Performance	Enter Marks	As a faculty member, I want to enter marks so that student performance is recorded.	High	5
US-18	Performance	Update Marks	As a faculty member, I want to update marks so that corrections can be made.	Medium	3
US-19	Performance	Calculate	As the system,	High	5

		Grades	I want to calculate grades so that results are generated consistently.		
US-20	Performance	Generate Result Sheet	As an authorized user, I want to generate result sheets so that academic results can be communicated.	Medium	5
US-21	Notifications	Attendance Alerts	As a student, I want attendance alerts so that I am informed about attendance status.	Medium	3
US-22	Notifications	Examination Notifications	As a student, I want examination notifications so that I know important examination information.	Medium	3
US-23	Notifications	Result Notifications	As a student, I want result notifications so that I know when results are available.	Low	2
US-24	Reporting	Student Reports	As an authorized user, I want student reports so that student information can be reviewed.	Medium	3
US-25	Reporting	Performance Reports	As an authorized user, I want	Medium	5

			performance reports so that academic performance can be reviewed.		
US-26	Reporting	Course Reports	As an authorized user, I want course reports so that course information can be reviewed.	Low	3

6. Acceptance Criteria

US-01 – Login

- Login page loads successfully.
- Registered users can log in.
- Invalid credentials display an appropriate error message.
- Successful login redirects to the dashboard.
- Session management functions correctly.

US-05 – Add Student

- Authorized user can open the add-student form.
- Required student information can be entered.
- Valid data is saved successfully.
- Invalid/duplicate data is handled appropriately.
- The new student appears in search/profile views.

US-06 – Search Student

- User can enter a search value.
- Matching student records are displayed.
- No-match cases are handled clearly.
- Selected student details can be opened.

US-14 – Record Attendance

- Faculty can select a course.
- Faculty can select a lecture date.
- Faculty can mark attendance.
- Attendance records are saved successfully.

- Attendance reports can be generated.

US-17 – Enter Marks

- Authorized faculty can select the student/course.
- Marks can be entered within defined limits.
- Valid marks are saved successfully.
- Invalid values show an error.
- Saved marks appear in the academic record.

US-19 – Calculate Grades

- System uses the defined grading rule.
- Grade is calculated from valid marks.
- Result is stored/displayed correctly.
- Invalid or incomplete marks are handled appropriately.

US-24 – Student Reports

- Authorized user can select report criteria.
- Relevant student data is displayed.
- Report can be generated successfully.
- Unauthorized users cannot access restricted data.

7. Prioritized Product Backlog

Prioritization is based on business value, user importance, technical dependencies and implementation complexity.

Priority	ID	Epic	Backlog Item	Points	Sprint
P1	US-01	Authentication	User Login	3	Sprint 1
P2	US-05	Student Management	Add Student	3	Sprint 2
P3	US-06	Student Management	Search Student	2	Sprint 2
P4	US-14	Attendance	Record Attendance	5	Sprint 3
P5	US-17	Performance	Enter Marks	5	Sprint 4
P6	US-04	Authentication	Role-Based Access Control	5	Sprint 1
P7	US-07	Student Management	Update Student	3	Sprint 2
P8	US-09	Student Management	View Student Profile	2	Sprint 2
P9	US-10	Course Management	Add Course	3	Sprint 3

P10	US-12	Course Management	Assign Students	5	Sprint 3
P11	US-15	Attendance	View Attendance	3	Sprint 3
P12	US-16	Attendance	Attendance Report	5	Sprint 3
P13	US-19	Performance	Calculate Grades	5	Sprint 4
P14	US-20	Performance	Generate Result Sheet	5	Sprint 4
P15	US-11	Course Management	Update Course	2	Sprint 5
P16	US-13	Course Management	View Course Details	2	Sprint 5
P17	US-24	Reporting	Student Reports	3	Sprint 4
P18	US-25	Reporting	Performance Reports	5	Sprint 4
P19	US-03	Authentication	Password Reset	3	Sprint 5
P20	US-21	Notifications	Attendance Alerts	3	Sprint 5
P21	US-22	Notifications	Examination Notifications	3	Sprint 5
P22	US-18	Performance	Update Marks	3	Sprint 4
P23	US-08	Student Management	Delete Student	3	Sprint 2
P24	US-26	Reporting	Course Reports	3	Sprint 5
P25	US-23	Notifications	Result Notifications	2	Sprint 5
P26	US-02	Authentication	Logout	1	Sprint 1

8. Story Point Estimation

- 1 – Very Easy
- 2 – Easy
- 3 – Moderate
- 5 – Difficult
- 8 – Very Difficult
- 13 – Highly Complex

Total estimated story points in the prepared backlog: 87. Planning Poker is used for collaborative estimation.

9. Sprint Planning

Sprint 1

Sprint Goal: Authentication and project setup

Sprint Backlog:

- Repository initialization
- User registration
- Login functionality
- Basic user interface
- Testing

Stories: US-01, US-02, US-04

Sprint 2

Sprint Goal: Student Management

Sprint Backlog:

- Add Student
- Search Student
- Update Student
- Delete Student
- Student Profile

Stories: US-05, US-06, US-07, US-08, US-09

Sprint 3

Sprint Goal: Course and Attendance Management

Sprint Backlog:

- Add Course
- Assign Students
- Record Attendance
- View Attendance
- Attendance Reports

Stories: US-10, US-12, US-14, US-15, US-16

Sprint 4

Sprint Goal: Academic Performance and Reporting

Sprint Backlog:

- Enter Marks
- Update Marks
- Calculate Grades
- Generate Result Sheets
- Student/Performance Reports

Stories: US-17, US-18, US-19, US-20, US-24, US-25

Sprint 5

Sprint Goal: Notifications and refinement

Sprint Backlog:

- Password Reset
- Course updates/details
- Attendance Alerts
- Examination Notifications
- Result Notifications
- Course Reports
- Final testing and improvements

Stories: US-03, US-11, US-13, US-21, US-22, US-23, US-26

10. Scrum Board

Backlog	To Do	In Progress	Testing	Completed
Password Reset	Add Student	Search Student	Attendance Report	Project Setup
Notifications	Record Attendance	Enter Marks	Login Testing	User Registration
Course Reports	Update Student	Calculate Grades	Student Search Testing	Basic UI

Update the Scrum Board throughout Sprint execution as work moves from Backlog to Completed.

11. Scrum Ceremonies

Sprint Planning

Select User Stories → Estimate effort → Define Sprint Goal → Create Sprint Backlog.

Daily Scrum

- What did I complete yesterday?
- What will I do today?

- What challenges am I facing?

Sprint Review

Demonstrate completed software features, verify acceptance criteria and collect feedback.

Sprint Retrospective

- What went well?
- What challenges occurred?
- What improvements should be implemented?

12. Agile Framework Comparison

Framework	Focus	Suitability	Reason
Scrum	Time-boxed Sprints and Scrum roles/ceremonies	High	Clear roles, backlog, sprints, reviews and retrospectives fit the project.
Kanban	Continuous flow of work	Medium	Useful for visualizing tasks, but the assignment emphasizes Scrum activities.
Extreme Programming (XP)	Engineering practices and frequent feedback	Medium	Useful for testing and code quality with iterative development.
Lean Development	Waste reduction and value delivery	Medium	Helpful for keeping the MVP focused and reducing unnecessary work.
Large Scale Scrum (LeSS)	Scaling Scrum	Low	More suitable for multiple teams at larger scale.
Scaled Agile Framework (SAFe)	Enterprise-scale Agile	Low	More process and coordination than required for a small student project.
Scrum@Scale	Scaling Scrum across teams	Low	Designed for coordination across multiple Scrum teams; unnecessary for a small team.

Recommended framework: Scrum, because the project maps naturally to Product Backlog, Sprints, Scrum roles, Daily Scrum, Sprint Review and Sprint Retrospective.

13. Burndown Chart

The accompanying Excel workbook contains a sample burndown dataset and chart. It tracks remaining story points over a sprint.

Day	Remaining Points
1	20
2	16
3	12
4	8
5	4
6	0

14. Sprint Review Report

The team demonstrates completed stories, checks the acceptance criteria, records stakeholder feedback and updates the Product Backlog. Incomplete items return to the backlog for re-prioritization.

- Completed features demonstrated
- Acceptance criteria checked
- Feedback collected
- Backlog updated

15. Sprint Retrospective Report

Area	Observation
What went well	Clear backlog priorities, defined acceptance criteria, collaborative estimation and a visible Scrum Board.
Challenges	Limited development time, dependencies between modules, testing effort and coordination.
Improvements	Break large stories into smaller tasks, improve daily communication, test earlier and keep the MVP focused.

16. Project Timeline and Release Roadmap

Phase	Focus	Main Deliverables	Release
Phase 1	Foundation	Project setup, authentication, basic UI	Release 1
Phase 2	Student Management	Student CRUD/search/profile	Release 2
Phase 3	Course & Attendance	Course assignment,	Release 3

		attendance and reports	
Phase 4	Performance & Reporting	Marks, grades, result sheets and reports	Release 4
Phase 5	Notifications & Refinement	Notifications, final testing and improvements	Release 4 refinement

17. Complete Deliverables Checklist

- ☐ Agile Project Planning Report
- ☐ Product Vision Document
- ☐ Scrum Team Structure
- ☐ Scrum Roles and Responsibilities Document
- ☐ Epic Documentation
- ☐ User Story Repository
- ☐ Acceptance Criteria Document
- ☐ Prioritized Product Backlog
- ☐ Story Point Estimation Sheet
- ☐ Sprint Planning Document
- ☐ Sprint Backlog
- ☐ Scrum Board
- ☐ Burndown Chart
- ☐ Sprint Review Report
- ☐ Sprint Retrospective Report
- ☐ Agile Framework Comparison Report
- ☐ Project Timeline and Release Roadmap
- ☐ Complete Agile Documentation

Conclusion

This Unit-2 Agile mini-project converts the Student Record and Academic Management Platform into a structured Agile delivery plan covering the required Unit-2 activities and deliverables.